

ProMS: An Integrated Computational Framework for Molecular Surface Analysis and Atomic-Level Hydrophobicity Mapping

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McGill University, Montreal

January 2026

A thesis submitted to McGill University in partial fulfillment of the requirements of the degree of
Master of Science (M. Sc.)

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List of Abbreviations

ALOGP – Atomic LogP

BAD – Biomolecular Adsorption Database

CLOGP – Calculated LogP

DGWIF – Wimley–White Interface Scale

DGOCT – Octanol–Water Partition Scale

GAFF – Generalized Amber Force Field

GUI – Graphical User Interface

IP – Isoelectric Point

kd – Dissociation Constant

LogP – Logarithm of the Partition Coefficient

MD – Molecular Dynamics

ML – Machine Learning

PDB – Protein Data Bank

ProMS – Protein Molecular Surface Calculator

QSAR – Quantitative Structure–Activity Relationship

SAS – Solvent-Accessible Surface

SES – Solvent-Excluded Surface

Abstract

Molecular surfaces, defined as the boundary between a molecule and its surrounding environment, play a central role in several key biological processes. The physicochemical features of these surfaces, such as charge, amphiphilicity, and geometric shape, determine how biomolecules interact, recognize binding partners, and assemble into complexes. Although there are existing molecular surface visualization tools, these are often cumbersome, inflexible, and unable to map properties at atom-level resolution. Here is presented ProMS (Protein Molecular Surface calculator), a modular Python-based software tool that unifies the Connolly analytical molecular surface with both force field-specific hydrophobicity schemes and a universal logP-derived atomic hydrophobicity scale. ProMS, through its leveraging of PyMOL, enables visualization of surface amphiphilicity and polarity at both atomic and residue resolution, and can be used to analyze proteins, nucleic acids, biomolecular complexes, and other small molecules. Its utility is presented through various case studies, in which ProMS analyzed globular proteins, visualized the inside of an ion channel, compared the surfaces of three steroids, and explored the interface of an aptamer-protein complex. Additionally, a thorough study of protein adsorption was conducted through the creation of a protein database and a subsequent machine learning analysis of their surface properties. By bridging experimental hydrophobicity data with analytical surface modeling, ProMS provides a flexible and accessible framework for studying molecular recognition and guiding rational design in structural biology, biotechnology, and drug discovery.

Résumé

Les surfaces moléculaires, définies comme la frontière entre une molécule et son environnement, jouent un rôle central dans plusieurs processus biologiques clés. Les caractéristiques physicochimiques de ces surfaces, telles que la charge, l'amphiphilicité et la forme géométrique, déterminent la manière dont les biomolécules interagissent, reconnaissent des partenaires de liaison et s'assemblent en complexes. Bien qu'il existe des outils de visualisation de surfaces moléculaires, ceux-ci sont souvent lourds, peu flexibles et incapables de cartographier des propriétés à une résolution atomique. Le présent travail introduit ProMS (Protein Molecular Surface calculator), un outil logiciel modulaire développé en Python qui unifie la surface moléculaire analytique de Connolly avec des schémas d'hydrophobicité spécifiques aux champs de force, ainsi qu'une échelle d'hydrophobicité atomique universelle dérivée du logP. En s'appuyant sur PyMOL, ProMS permet de visualiser l'amphiphilicité et la polarité de surface à la fois à l'échelle atomique et à l'échelle des résidus, et peut être utilisé pour analyser des protéines, des acides nucléiques, des complexes biomoléculaires et d'autres petites molécules. Son utilité est illustrée au moyen de plusieurs études de cas dans lesquelles ProMS a analysé des protéines globulaires, visualisé l'intérieur d'un canal ionique, comparé les surfaces de trois stéroïdes et exploré l'interface d'un complexe aptamère-protéine. De plus, une étude approfondie de l'adsorption protéique a été réalisée grâce à la création d'une base de données de protéines et à une analyse d'apprentissage automatique de leurs propriétés de surface. En reliant des données expérimentales d'hydrophobicité à une modélisation analytique des surfaces, ProMS constitue un cadre flexible et accessible pour l'étude de la reconnaissance moléculaire et pour guider la conception rationnelle en biologie structurale, en biotechnologie et dans la découverte de médicaments.

Acknowledgements

I would like to sincerely thank my supervisor, Professor Dan V. Nicolau, for his tireless guidance, support, and encouragement throughout my master's thesis and during our broader collaboration. Professor Nicolau's expertise, insight, and enthusiasm have been invaluable in shaping this project and in helping me grow both as a researcher and as a person. I am also deeply grateful to the members of Professor Nicolau's laboratory for their technical contributions, stimulating discussions, and camaraderie over the past few years. On a personal note, I would like to thank my friends and family for their unwavering support, patience, and positivity. This academic journey would not have been possible without the diverse perspectives, mentorship, and kindness I have encountered along the way.

Contribution of Authors

All chapters presented in this thesis were written by Matthew Cho with editing and revision notes from Professor Dan V. Nicolau. Matthew Cho designed and implemented the ProMS application and led the case studies and analysis discussed in this document. Matthew Cho created all the figures presented in this thesis. Kevin Xu and Angela Wang assisted with the sections and analyses pertaining to ion channels, atomic hydrophobicity derivations, and steroids. Zahra Mahmoodi, Prasad Shetty, and Lauren Harrison were responsible for the creation of the BAD database entries, on which Matthew Cho performed additional software upgrades and subsequent machine learning analyses. OpenAI's ChatGPT was used after the original draft was written to improve the flow and clarity and to translate the abstract to French.

1. Introduction

1.1. Molecular Surfaces

The understanding of biomolecular recognition, which is the cornerstone of biological processes at the molecular scale, has evolved from the early induced-fit hypothesis toward a more dynamic conformational selection mechanism [1]. While the internal architecture of biomolecules, particularly proteins with their intricate secondary-structure motifs such as α -helices, β -sheets, and coils, has received extensive scientific attention, culminating in large-scale structural databases, most biomolecular interactions occur at the molecular interface [2]. The molecular surface, defined as the boundary between a molecule and its surrounding environment, plays a central role in such recognition processes [1, 3]. The physicochemical features of this surface govern key biological functions, including receptor-ligand binding, enzyme catalysis, and macromolecular complex formation [4]. Consequently, molecular surface analysis has become relevant to a broad range of disciplines, with important applications in biotechnology, pharmaceutical development, and materials science [5, 6].

Despite their central importance, molecular surfaces have historically received less academic attention than the internal structures of biomolecules. For example, the Protein Data Bank (PDB), which stores experimentally determined and computationally modeled protein structures, was established more than half a century ago and now contains nearly 250 000 experimental and over 1 000 000 calculated entries [2]. In contrast, protein and nucleic acid molecular surfaces are typically computed on demand from PDB data and deposited in much more recent, specialized repositories [7]. This disparity in focus partly stems from the conceptual and computational complexity of defining and analyzing biomolecular surfaces. Their description depends on several interrelated physicochemical features, including geometric shape, electrostatic potential, and

hydrophobicity, which determine how a molecule interacts with its surroundings and recognizes complementary partners [8]. Collectively, these properties govern the affinity and specificity of biomolecular recognition events [9]. By quantitatively characterizing and visualizing these surface features, researchers can more directly link molecular structure to function, which is an essential step in drug discovery, biomaterials development, and protein engineering.

The most apparent aspect of a biomolecular surface is its geometrical boundary, which defines the overall shape of the molecule. Surface curvature and topology modulate molecular complementarity and steric fit, influencing how biomolecules interact in three-dimensional space [10]. Several computational approaches, such as the Connolly surface, Sanner's reduced surface, and Gaussian-, dot-, and mesh-based representations have been developed to analytically describe molecular surfaces [11, 12]. Among these, the Connolly molecular surface algorithm remains the gold standard for defining the geometric boundary of a molecule [13]. The algorithm constructs the surface by rolling a virtual probe sphere over the molecule's atoms at a predefined distance from their center. This probe radius is often, but not exclusively, chosen to be the atomic van der Waals radius. In this configuration, the generated surface corresponds to the solvent-excluded surface (SES), which is composed of exposed atomic spheres, toroidal regions where the probe contacts two atoms simultaneously, and re-entrant cavities formed by multiple-atom contacts [14]. Connolly's molecular surface program can also output the solvent-accessible surface (SAS), the van der Waals surface, and the solvent-excluded volume [13]. The Connolly algorithm is highly regarded for its ability to generate smooth, continuous, and analytically precise surfaces that accurately reflect molecular geometry and solvent accessibility. However, its computational complexity, dependence on probe radius selection, and limited ability to directly incorporate

physicochemical surface properties, such as electrostatics or hydrophobicity, pose challenges for large systems or for fully integrated analyses.

As the functional behavior of a biomolecule is also determined by the spatial distribution of its physicochemical properties, the second, more complex aspect of the molecular surface concerns the external distribution of charges. Although individual charges are localized on individual atoms, the long-range nature of electrostatic interactions gives rise to an all-atoms-involved electrostatic potential field [7]. The atoms positioned on or near the geometrical molecular surface exert a disproportionately strong influence on the external electrostatic potential, since inner atomic contributions are increasingly shielded by the surrounding charge distribution [15]. Because the exact computation of molecular electrostatic potentials is often non-trivial and computationally demanding, the spatial arrangement of charged surface atoms can serve as a useful first-order approximation of the molecule's overall electrostatic character [16].

Electrostatic interactions mediate the initial attraction and orientation between biomolecules through polar and ionic forces [15]. Owing to their long-range and reversible nature, these interactions primarily function as steering mechanisms, as in they guide molecular partners toward one another rather than locking them into a stable complex. Hydrophobic interactions, by contrast, arise from a fundamentally different physical origin. Instead of stemming from attractive or repulsive forces, hydrophobicity reflects differences in the Gibbs free energy of a system, such as an atom, functional group, or surface region, when water molecules are present or excluded [17]. As a result, hydrophobic interactions act over very short distances, typically on the order of one or two water-molecule diameters [18]. Moreover, because they represent an underlying free-energy minimization process, hydrophobic interactions are largely irreversible, in contrast to the dynamic and reversible nature of electrostatics. Consequently, hydrophobicity plays a crucial role in

stabilizing molecular complexes once electrostatic forces have brought the interacting partners into close proximity [19].

1.2. Hydrophobicity Mapping

While hydrophobicity has long been recognized as a central determinant of biomolecular stability and interaction, its quantitative representation at the atomic level remains a complex and evolving field [17]. Traditional hydrophobicity scales, including those based on experimentally measured free energies such as the Wimley–White interface scale (DGWIF) and the octanol–water partition scale (DGOCT), characterize hydrophobic tendencies at the residue level, but do not capture the nuanced distribution of amphiphilicity across individual atoms within a biomolecule [20, 21]. More recent studies have introduced atomic-resolution hydrophobicities by assigning hydrophobicity values to force-field–derived atom types; however, these approaches are largely restricted to proteins, limiting their applicability to broader classes of biomolecules [22, 23]. To enable comparisons across proteins, nucleic acids, and small molecules (e.g., for drug candidates), a universal atomic hydrophobicity scale is required. Fragment-based quantitative structure–activity relationship (QSAR) methods, including ALOGP (atomic logP) and CLOGP (calculated logP), provide such a framework by estimating the octanol–water partition coefficient (logP) through the summation of predefined fragment hydrophobic contributions, each representing the combined effects of the atoms and bonds within the fragment [24].

1.3. Protein Adsorption on Solid Surfaces

Protein adsorption is a ubiquitous biological process that plays a critical role in diverse contexts, including protein interactions with biological membranes and the formation of microbial biofilms

[25, 26]. The adsorption of proteins onto abiotic solid surfaces is equally important across numerous applications. These include medical and dental practices (e.g., implants and catheters), biomedical research (e.g., drug delivery and controlled release), and diagnostic or drug-discovery technologies such as assays, microarrays, and lab-on-a-chip devices [27-33]. Importantly, protein adsorption may be beneficial or detrimental depending on the specific context. For example, high-throughput proteomic microarrays require both strong immobilization of proteins on the chip surface and preservation of their native conformation, which are two requirements that often conflict [34]. Nanoparticles can serve as efficient carriers for intracellular protein delivery in therapeutic settings, yet they may also induce toxicity through unintended inactivation of essential proteins [35-37]. Likewise, protein adsorption on biomaterials can be undesirable if it triggers an immune response, but it is central to tissue engineering, where it modulates cell activation, adhesion, and wound healing [38, 39].

The ubiquity and importance of protein adsorption have led to an extensive body of research; for instance, more than 3 000 articles include “protein adsorption” in the title, and over 12 000 list it as a keyword. Nevertheless, despite this substantial literature, it remains impossible to reliably predict how much of a given protein will adsorb onto a specific surface from a given fluid medium [40]. Even less is known about the structural integrity or bioactivity of the protein following adsorption, underscoring a major gap in our current understanding.

1.4. Research Objectives

Despite the availability of robust molecular visualization platforms, no existing tools can provide atomic-level hydrophobicity mapping with analytical surface computation. Current workflows

often rely on separate software for surface generation, physicochemical property assignment, and visualization, thus creating barriers to reproducibility, accessibility, and ease of use.

In response to these challenges, the objective of this work is to develop and validate ProMS (Protein Molecular Surface calculator), a modular Python-based plugin for PyMOL designed to unify these tasks into a single, cohesive framework [41]. Specifically, this research aims to:

1. Implement a streamlined system capable of computing molecular surfaces at both atomic and residue-level granularity.
2. Integrate hydrophobic, electrostatic, and geometric surface descriptors using both force field-specific hydrophobicity schemes and a universal logP-derived atomic scale.
3. Evaluate the performance and accuracy of ProMS through comparative hydrophobicity analyses.
4. Demonstrate ProMS' versatility across proteins, aptamers, small molecules through targeted case studies.
5. Leverage ProMS and machine learning algorithms to aid in predicting and understanding protein adsorption on solid surfaces.

By achieving these objectives, this work establishes ProMS as a comprehensive platform for studying biomolecular recognition, interface complementarity, and solvation phenomena with atomic-level resolution.

2. Methods

2.1. Connolly Algorithm

ProMS uses the Connolly algorithm to generate the analytical molecular surface. Connolly has made his surface algorithm available in a package written mostly in C [42]. This package includes an executable named “msRoll”, which allows the Connolly algorithm to be ran on PDB files. The user can specify the radius size of the probe sphere, and the executable outputs files related to the atomic areas and volumes of the target molecule. The data generated from msRoll is later used in downstream hydrophobicity and molecular surface property calculations.

2.2. Python

ProMS was implemented in Python, chosen for its readability, extensive scientific ecosystem, and straightforward integration with external applications [43]. Within Python, ProMS makes extensive use of the scientific libraries NumPy and Pandas for numerical and tabular data processing [44, 45]. NumPy provides efficient array operations for handling atomic coordinates, surface meshes, and per-atom or per-residue properties, while Pandas is used to organize, filter, and export hydrophobicity, charge, and surface-area descriptors in a structured tabular format. The graphical user interface (GUI) was implemented using PySide and Qt, which enable event-driven interaction, parameter input, and dynamic visualization control within a cross-platform environment [46]. Finally, ProMS leverages PyMOL’s Python API to access molecular structures, manipulate representations, and render molecular surfaces directly in the PyMOL viewer, ensuring a tight coupling between analytical routines and high-quality molecular graphics [41].

2.3. Schrodinger's PyMOL

PyMOL is an open-source molecular visualization system maintained and distributed by Schrodinger [41]. It is widely used in the domain of molecular modeling due to its inherent versatility and accessibility [47, 48]. In brief, PyMOL allows the user to analyze and visualize complex 3-dimensional molecular structures. Additionally, it provides comprehensive tools to generate high-quality images and movies. The major upside of PyMOL is its Python backbone. This allows researchers to write scripts and workflows to automate tasks and perform custom analyses. PyMOL also encourages the creation of plugins, which can easily be shared within the user community. ProMS is one of these plugins and extends PyMOL directly.

2.4. Force Field-Based Atomic Hydrophobicities

To obtain high-resolution hydrophobicity representations on protein molecular surfaces, residue-level hydrophobicity scales were decomposed into atomic hydrophobicities, as proposed by Nicolau et al. in 2014 [22]. Two well-established residue-based hydrophobicity scales were used as references, DGWIF and DGOCT [49]. In both scales, positive values indicate hydrophilicity, and negative values indicate hydrophobicity.

Each amino acid's hydrophobicity for the given hydrophobicity scale was expressed as the sum of contributions from its constituent atom types, as seen in equation 1 [22].

$$(1) \quad \sum_{i=1}^m AtHypho_i \cdot n_{ij} = AAHypho_j$$

Where:

i = atom type index

j = amino acid index (1–20)

n_{ij} = number of atoms of type i in amino acid j

AtHypho_i = atomic hydrophobicity of atom type i

AAHypho_j = hydrophobicity of amino acid j (from residue scale)

This yields a system of 20 equations, one for each amino acid, which was solved to obtain atomic hydrophobicity values consistent with the original residue scale. Atom types were assigned using the General AMBER Force Field (GAFF) and BIO++ [50, 51]. These force fields distinguish between chemically distinct but structurally similar atoms (e.g., sp² vs. sp³ carbons), allowing more precise hydrophobicity assignment. Backbone atoms were not treated separately, as they are represented naturally in the atom-type classification.

2.5. Universal Atomic Hydrophobicity Scale

To enable atomic hydrophobicity mapping for non-protein molecules (e.g., nucleic acids, ligands, small molecules), a universal scale was developed using fragmental logP approaches from the ALOGP and CLOGP methods [24]. These fragmental contributions were mapped to corresponding atom types, producing a universal set of atomic hydrophobicity values applicable across diverse molecular classes. In this scale, negative values indicate hydrophobicity, and positive values indicate hydrophilicity, consistent with the protein-specific scales. The universal scale assigns hydrophobicity to atom types including aliphatic carbons, aromatic carbons, heteroatoms, and charged groups.

2.6. BAD Database

The Biomolecular Adsorption Database (BAD 2.0), developed as an upgraded and expanded version of the original BAD 1.0, is an archive of experimentally reported data on protein adsorption onto flat solid surfaces [32, 52]. Designed as a web-oriented, open-access resource, BAD 2.0 includes only entries that provide complete quantitative descriptors for the three primary classes of parameters relevant to protein adsorption:

1. Protein descriptors, ideally with structures identifiable in the Protein Data Bank (PDB), enabling subsequent extraction of molecular and surface properties [2].
2. Surface descriptors, including surface type as well as wettability parameters such as water contact angle or surface tension.
3. Fluid-phase descriptors, including protein concentration, pH, ionic strength, and temperature.

In addition to these core variables, BAD 2.0 stores auxiliary metadata, such as the experimental method used to quantify adsorption and the DOI of each source publication, facilitating rapid retrieval and verification of the original data.

BAD 2.0 is publicly available at <https://www.bionanoinfo.com/bad/> and is designed for continuous expansion, either through scheduled updates or community-submitted entries vetted by the database maintenance team. A screenshot of the BAD 2.0 entry portal is provided in **Figure 1**, and the entity–relationship diagram describing the database architecture appears in **Figure 2**. At the time of the present analysis, BAD 2.0 contains 865 curated protein adsorption records.

10 ▾ entries per page

Search:

ID ▴ ▾	Protein ▴ ▾	PDB ▴ ▾	Surface Concentration ▴ ▾	Solution Concentration ▴ ▾	Adsorbing Surface ▴ ▾	Contact Angle ▴ ▾	Surface Tension ▴ ▾	pH ▴ ▾	Ionic Strength ▴ ▾	Temperature ▴ ▾
1	Human Serum Albumin	1BJ5	1.1000	0.0050	Gold	44.00	4.81	3.50	0.001	25.0
2	Human Serum Albumin	1BJ5	2.7000	0.0050	Gold	44.00	4.81	3.50	0.150	25.0
3	Human Serum Albumin	1BJ5	3.0000	0.0100	Gold	44.00	4.81	3.50	0.150	25.0
4	Fibrinogen	1M1J	1.2000	0.0012	Silica	36.00	2.59	3.50	0.010	20.0
5	Fibrinogen	1M1J	3.2000	0.0012	Silica	36.00	2.59	4.50	0.010	20.0
6	Fibrinogen	1M1J	4.1000	0.0012	Silica	36.00	2.59	7.40	0.010	20.0
7	Fibrinogen	1M1J	4.2000	0.0012	Silica	36.00	2.59	7.40	0.150	20.0
8	Fibrinogen	1M1J	5.5000	0.0100	Silica	36.00	2.59	7.40	0.150	20.0
9	Fibrinogen	1M1J	5.6000	0.0200	Silica	36.00	2.59	7.40	0.150	20.0
10	Fibrinogen	1M1J	5.7000	0.0400	Silica	36.00	2.59	7.40	0.150	20.0

Showing 1 to 10 of 860 entries

< 1 2 3 4 5 ... 86 >

Figure 1 – Screenshot of BAD2.0 Online Database. The database, along with other resources can be found at <https://www.bionanoinfo.com>.

The architecture of BAD 2.0 consists of a series of interlinked MySQL tables, displayed dynamically on web pages. Although the core BAD table is the primary focus of the present work, the database schema also includes tables describing protein properties (e.g., physicochemical attributes, structural identifiers) as well as computed molecular surface descriptors generated using ProMS. This relational structure minimizes redundancy, supports database integrity, and accommodates future expansion. All tables are optimized to be rapidly filterable, sortable, and searchable, enabling efficient extraction of targeted subsets of adsorption data. Moreover, a suite

of Python-based maintenance scripts was developed to support scalability, data validation, and automated integration of newly extracted or user-submitted entries.

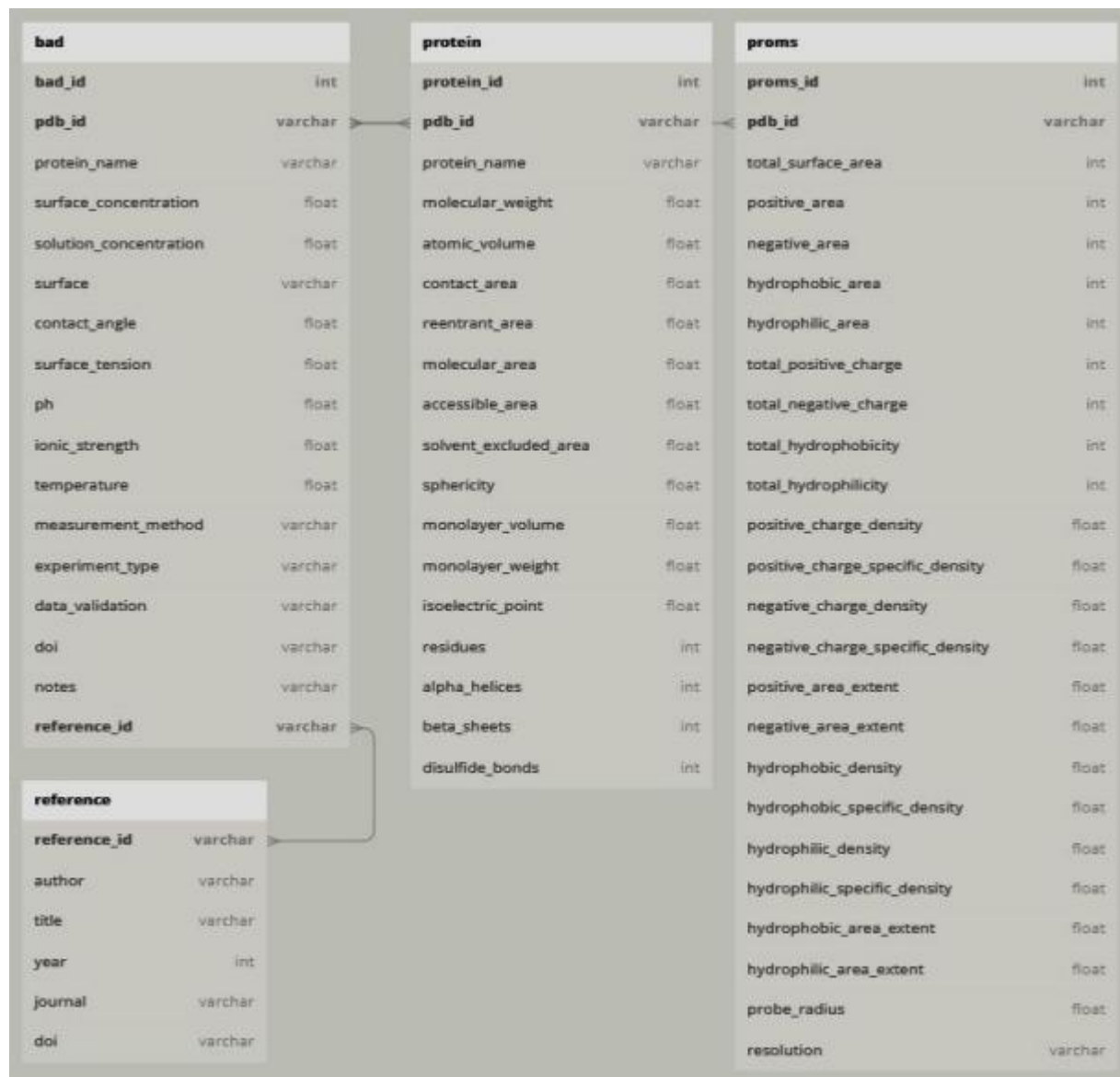


Figure 2 – Entity relationship diagram for BAD2.0. BAD2.0 contains information pertaining to the protein adsorption experimental measurements, as well as data about the protein itself calculated via ProMS.

2.7. Machine Learning

Machine learning techniques were employed to investigate the relative importance of molecular, surface, and solution-phase variables in predicting protein adsorption. All analyses were conducted in Python using Jupyter Notebook, a widely used platform in data science that enables the integration of executable code, explanatory text, and graphical outputs within a single environment [53]. Data manipulation and visualization were performed using Pandas, NumPy, matplotlib, and scikit-learn, which together provided a robust computational framework for exploratory analysis and model development [44, 45, 54, 55].

The complete curated dataset contained over 50 features, each corresponding to a distinct physicochemical, structural, or experimental descriptor of protein adsorption. The objective of the feature exploration phase was to identify the variables most strongly associated with protein adsorption and to eliminate features that were redundant or minimally informative. From the full dataset, two smaller subsets were constructed based on surface hydrophobicity: a hydrophobic subset comprising 111 adsorption records and a hydrophilic subset comprising 71 records. Within each subset, additional feature groups were generated, representing different combinations of explanatory variables used as inputs for adsorption prediction. In all cases, the dependent variable was the experimentally measured surface concentration of the adsorbed protein.

For each feature group, analyses were performed separately on the hydrophobic and hydrophilic datasets. All variables were first normalized to ensure comparability of scales. Feature importance was then explored using ridge regression, a regularized linear model commonly employed in feature selection pipelines [56]. Ridge regression was chosen for its ability to account for multicollinearity among variables while preventing overfitting through an L_2 penalty term. After

fitting the model, the resulting coefficient magnitudes were extracted and visualized to provide a qualitative measure of each variable's contribution to protein adsorption.

Because coefficient-based feature analysis does not directly quantify predictive accuracy, additional models were employed to assess performance. For each feature group, three regression models (ordinary least squares, ridge regression, and random forest regression) were trained and evaluated on both the hydrophobic and hydrophilic datasets (six models per feature group). To ensure robust estimation and to minimize model bias and variance, each model was evaluated using 5-fold cross-validation, after shuffling the dataset. Model performance was quantified using the average coefficient of determination (R^2) across the five folds, providing an interpretable measure of predictive ability.

3. Results & Discussion

3.1. Protein-Derived Atomic Hydrophobicities

Atomic hydrophobicities were derived from the matrix-based decomposition of residue-level hydrophobicity scales (DGWIF and DGOCT) using GAFF atom types [22]. To validate this approach, reconstructed residue hydrophobicities were compared to their original scale values. The reconstructed values displayed strong correlations with the original scales ($R^2 = 0.837$ for DGWIF and $R^2 = 0.7904$ for DGOCT), confirming that the decomposition preserved residue-level trends without overfitting. Notably, a perfect correlation was not the target here, as some deviation is desirable to capture additional atomic-level variation that residue-averaged scales inherently smooth out.

GAFF-derived atomic hydrophobicities were further compared to those previously calculated using BIO++ for both residue scales, as shown in **Figure 3** [23]. The correlation between GAFF and BIO++ values was stronger for DGOCT ($R^2 = 0.636$) than for DGWIF ($R^2 = 0.462$). Given that the two scales are based on different physicochemical models, perfect agreement was not expected. Compared to BIO++, GAFF has fewer unique atom types for the amino acids. This allows for higher generalizability and a more robust baseline with less overfitting.

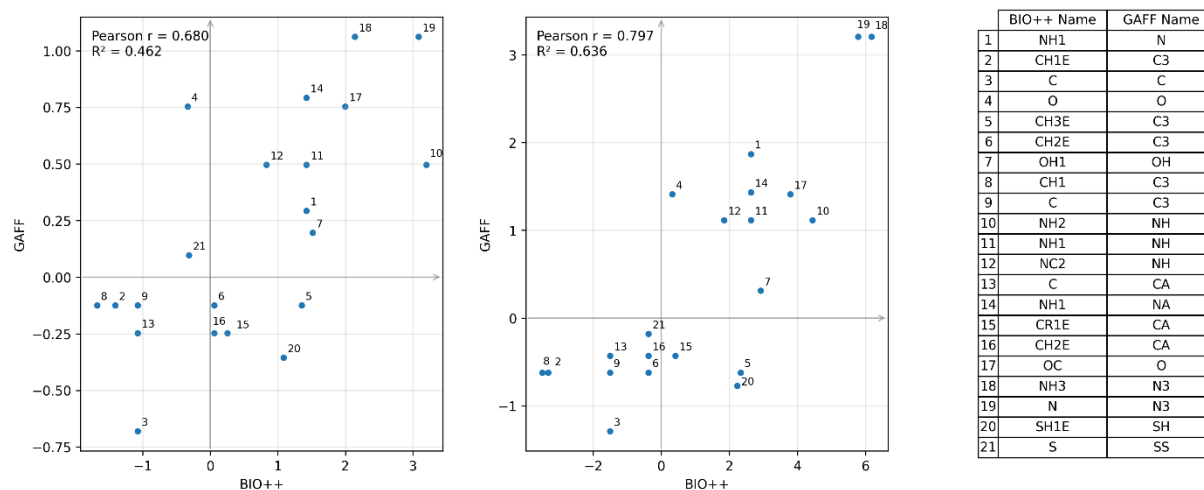


Figure 3 – Comparison of GAFF and BIO++ derived atomic hydrophobicities. Atomic hydrophobicities derived for DGWIF (left) and DGOCT (middle). Atom type comparison in table (right).

In both plots, the majority of points lie within the first ((+, +)) and third ((-, -)) quadrants. This indicates that atoms with positive (hydrophilic) or negative (hydrophobic) values in one force field generally retain the same amphiphicity in the other. This alignment across quadrants demonstrates a consistent directionality in hydrophobic character assignment, although with slightly different magnitudes. It should be noted, however, that this alignment is stronger for the DGOCT hydrophobicity scale.

The stronger GAFF–BIO++ agreement for DGOCT compared to DGWIF likely reflects differences in the underlying experimental models. Octanol–water partitioning (DGOCT) measures solvation tendencies more directly at the molecular level, which aligns well with atom-based hydrophobicity assignment [57]. In contrast, bilayer–water partitioning (DGWIF) incorporates additional factors such as membrane deformation and lipid packing effects, which may introduce noise when decomposed into atomic contributions [49].

While the protein-specific atomic hydrophobicity scales compared in **Figure 3** show a high degree of internal consistency, a direct quantitative comparison between these scales and the universal

logP-derived atomic hydrophobicity scale does not yield meaningful correlations. Indeed, preliminary analyses revealed very low coefficients of determination (R^2 values close to zero), which makes sense given the difference in experimental methodologies used in generating these scales. Protein-derived atomic hydrophobicity scales are obtained by decomposing experimentally measured free energies of transfer within the structural and chemical context of proteins. As such, these scales implicitly encode effects arising from protein-specific environments, including backbone constraints, secondary-structure packing, and cooperative interactions among neighboring residues.

In contrast, logP-derived atomic hydrophobicity values originate from fragment-based QSAR models designed to predict the octanol–water partition coefficient of small, freely solvated molecules. These models assign atomic contributions based on chemical fragments optimized to reproduce bulk partitioning behavior, without reference to macromolecular context, conformational restriction, or solvent-excluded environments typical of folded proteins. In other words, universal logP models employ a much broader and more finely partitioned atom-type space, designed to describe the full range of organic chemistry encountered in small molecules. This expanded atom-type vocabulary necessarily redistributes hydrophobic contributions so that they are shared across a wider range of atoms (not solely proteins). Because the two approaches differ both in their reference systems (proteins vs. small molecules) and in the physical meaning of the derived quantities, a low correlation is expected and does not imply inconsistency. Instead, it highlights the complementary nature of protein-derived and logP-based hydrophobicity scales, underscoring the value of logP as a universal descriptor applicable across molecular classes rather than as a direct replacement for protein-specific hydrophobicity measures.

3.2. Universal Atomic Hydrophobicity Scale Based on LogP

In this implementation, fragment contributions are translated into atomic-level hydrophobicity values. Each unique atom type, classified by hybridization state, functional group context, or aromaticity, is assigned a hydrophobicity coefficient derived from its fragmental contribution in the ALOGP/CLOGP models. This mapping ensures broad applicability: the same atomic parameters can be used across proteins, nucleotides, steroids, ligands, and other small molecules. This enables direct comparisons of surface hydrophobicity across structurally diverse systems.

Consistent with the sign conventions used in protein-specific scales, negative values denote hydrophobicity, while positive values denote hydrophilicity. For instance, aliphatic (methyl/methylene) fragment contributions are strongly negative (e.g., -1.56 for methyl, -1.01 for methylene), aligning with their well-known nonpolar characteristics. Aromatic carbons and heteroatoms receive values reflecting their intermediate or polar nature, as determined by empirical fitting in the original fragmental methods.

This universal hydrophobicity scale offers several advantages. First, it is physico-chemically grounded because it is derived from experimental partitioning data (logP), ensuring direct relevance to solvation phenomena and intermolecular interactions. Second, it provides granularity and consistency by assigning atom-level hydrophobicity values across a wide range of molecular types, enabling high-resolution mapping of hydrophobic and hydrophilic regions. Third, it offers flexibility because the use of fragment libraries allows straightforward extension to new atom types, providing a clear pathway for future plugin expansion and customization. By integrating both the protein-specific scales (DGWIF and DGOCT) and the “universal” hydrophobicity scale, ProMS delivers a robust and adaptable framework for atomic-level hydrophobicity visualization applicable across diverse molecular classes.

3.3. Protein Molecular Surface Calculator (ProMS)

The Protein Molecular Surface Calculator (ProMS) was first introduced in 2024 as a means to calculate various surface properties of proteins, with an emphasis on atomic amphiplicities [52]. More specifically, it is a downloadable application for the quantitative analysis and visualization of molecular surfaces. This software package comes pre-loaded with all the files needed to analyze proteins via the methodology involving atomic hydrophobicities derived from residue scales proposed by Nicolau et al. [22]. Originally developed as an upgrade to its predecessor PSpC (Protein Surface Property Calculator), this latest version of ProMS has expanded capabilities, with the most notable upgrade being its ability to process, analyze, and visualize a larger selection of molecules [23]. Through the integration of a “Universal Atomic Hydrophobicity Scale”, ProMS can natively support nucleic acids, ligands, and small organic molecules. It also has extra functionality and customizability and can now be used as both an analytical tool and a framework for creating researcher-specific workflows. The result is a simple-to-use, yet advanced software package with applications in structural biology, molecular modeling, protein-protein interactions, and drug discovery.

3.4. ProMS Architecture

ProMS was developed using a modular Python framework [43]. Each component of the application is responsible for a specific function, such as input handling, surface property calculation, visualization, output generation, and user interaction. This separation allows for easier code maintenance and readability. It also enables extensibility, allowing new analytical methods or visualization features to be incorporated as separate modules without disrupting existing

workflows. While the majority of ProMS was developed as an in-house program, it does make use of two external dependencies, Connolly's msRoll executable and Schrodinger's open source PyMOL [41, 42].

As schemed in **Figure 4**, ProMS architecture is structured in a few critical components. At the front-end, a graphical user interface (GUI) makes it easy for the user to select their desired parameters and upload any additional files, if needed. Built using PySide and QT widgets, the intuitive GUI also includes input validation and error handling to ensure smooth downstream computations. ProMS comes packaged with several index files which contain the data needed to perform the surface property calculations. This data has either been calculated elsewhere in-house or taken from reputable literature sources. However, these index files are easily editable and extendable, meaning the user can alter them or add custom values as they deem fit for their personal research needs.

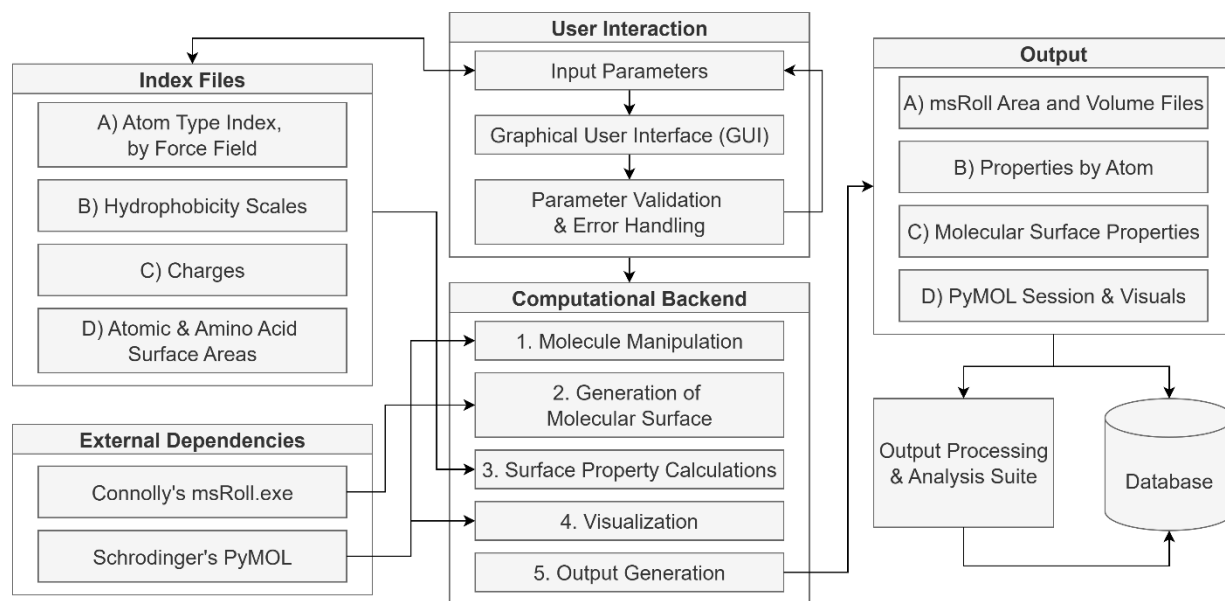


Figure 4 – ProMS Architecture. ProMS was built using a modular approach to ensure flexibility, extensibility, and maintainability. While most of the program was written in-house, PyMOL and Connolly's msRoll aid with some of the backend computations.

Once the inputs have been selected and verified, they are passed to the computational backend component of the application. This involves five sequential steps. Firstly, the molecule's PDB file is preprocessed to ensure proper formatting, and unwanted atoms (hydrogens, solvents) are removed. Next, Connolly's msRoll executable is run on the processed PDB file to generate the molecular surface. This outputs a file which includes the exposed area of each atom which comprises the surface. Thirdly, molecular surface properties are calculated using the methodology proposed by Nicolau et al. [22]. These properties are then used for the visualization, which is done in PyMOL. Lastly, all the generated data is cleaned up and organized. It gets output locally and there is also the option to connect to an SQL database for data aggregation purposes.

3.5. ProMS Features

ProMS allows for highly customizable analysis through parameter selection. Parameters can easily be chosen from the intuitive graphical user interface that comes pre-installed with the application. There are five main input parameters necessary for a molecular surface analysis: the molecule of interest, the probe sphere radius, the pH, the force field, and the hydrophobicity scale. ProMS comes equipped with input validation and error handling to ensure that the downstream calculations run seamlessly. A snapshot of the GUI can be seen in **Figure 5**.

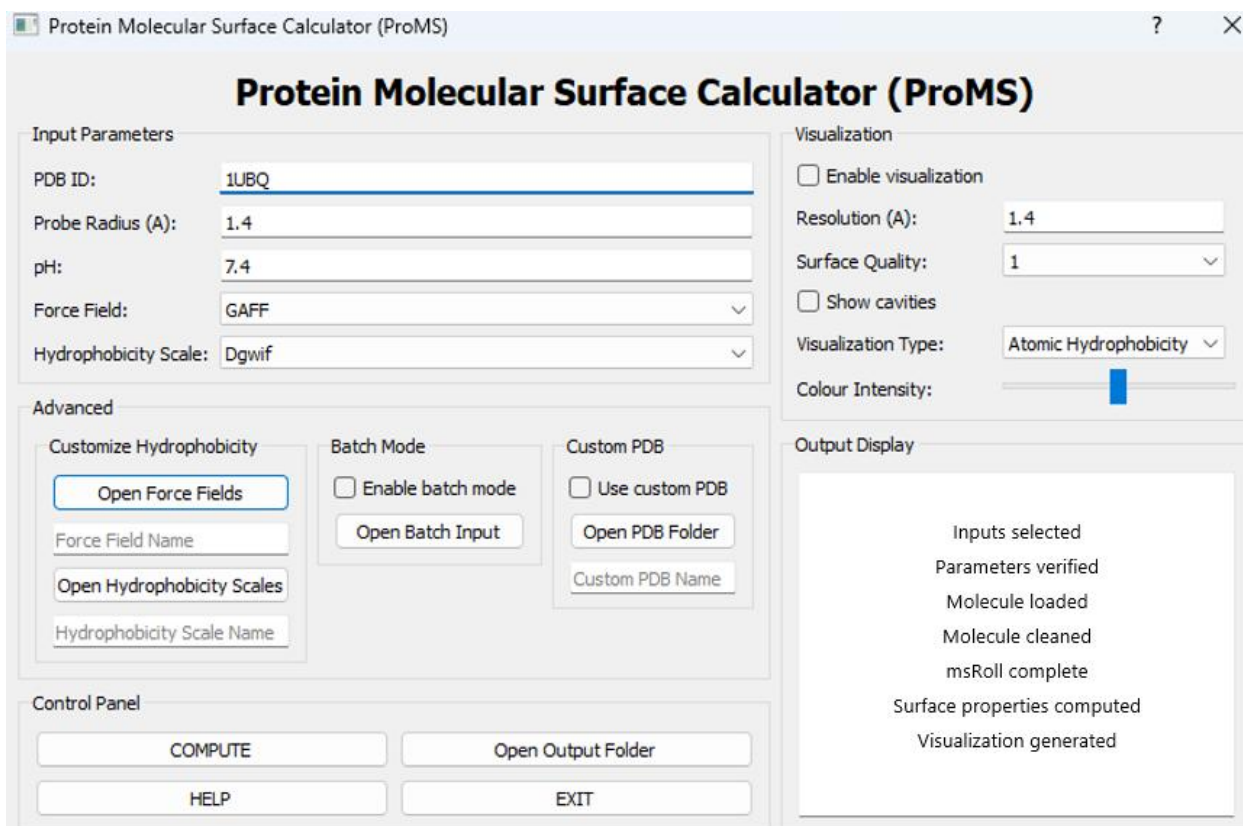


Figure 5 – ProMS’ Graphical User Interface (GUI). ProMS’ GUI makes it easy for the user to select their desired parameters and contains all the necessary information needed to perform a surface property analysis. ProMS integrates itself seamlessly with Schrodinger’s PyMOL, where it can be installed as a plugin.

ProMS has atom types for proteins and nucleic acids natively embedded. The user can choose to analyze a molecule from an online database via its PDB ID or can provide a custom PDB file that they have created or modified themselves [2]. Additionally, custom atom types can be easily defined in-app, allowing for even more flexibility. Essentially, ProMS offers a framework to characterize any type of molecule, however the program must be provided with a correctly formatted PDB file [58].

The probe radius is a very important parameter, as it directly affects the generation of the molecular surface via Connolly’s algorithm [14]. It refers to the radius of the probe sphere that is rolled over the van der Waals surface of the molecule to map out the boundary accessible to the solvent.

Consequently, varying the probe radius directly influences both the geometry and the derived properties of the surface. A smaller probe radius penetrates deeper into surface grooves and cavities, producing a more detailed and higher resolution surface. Conversely, a larger probe radius smooths over fine features, bridges narrower crevices, and reduces total surface area. It should be noted that the larger the probe radius, the longer the msRoll executable takes to run to completion.

The pH and selected force fields and hydrophobicity scales are related to the surface property calculations. The pH will affect the charge distribution, whereas the selected force field and hydrophobicity scale quantify the molecule's atomic and amino acid (if the target molecule is a protein) amphilities, respectively. ProMS comes packaged with several "index files" that contain the numerical data values for these parameters. However, as is the theme for the newest version of ProMS, these index files can be edited or appended to, allowing for fully customizable analyses.

ProMS uses PyMOL as a molecular manipulation and visualization tool [41]. PyMOL is used firstly to remove any unwanted atoms (i.e., hydrogens, solvent molecules) that will hinder the surface property calculations. Given the nature of molecular surfaces, a good visualization tool is highly necessary. ProMS allows for the molecule to be colored according to charge or hydrophobicity (both atomic and amino acid resolution), with a gradient showing the relative intensity of the property. There are several built-in visualization parameters, including the surface resolution and quality, presence of cavities, and color intensity. Furthermore, ProMS automatically creates a high-resolution image and a 360-degree short video of the visualized molecule.

ProMS comes equipped with "Batch Mode", which is a methodology that allows for many characterizations to be run consecutively without the need for user-intervention. This makes it more convenient to perform larger analyses, as the program can run in the background without the need to enter the desired parameters for each individual computation. Additionally, ProMS

provides the option to connect to a database (SQL-type), which allows for smoother external numerical processing and profiling [59]. In whole, each ProMS computation outputs several files: two Connolly executable raw outputs (atomic area and volume), a list of atomic hydrophobicities, the molecule's surface properties, and files pertaining to the visualization (PyMOL session, .tiff image, and .mp4 movie).

ProMS is a powerful tool that can be used as is out of the box, or “hijacked” to fit the users' research needs. Foremost, it specializes in protein analysis as it has tailored methodology and protein-derived atomic hydrophobicities built-in to the backend. However, its powerful visualization module and customizable inputs make it a flexible jackknife applicable to many research areas. Special care has been put into writing scalable code, so that new modules can easily be added without the need to alter existing software. Excitingly, ProMS' integration into PyMOL allows for seamless use of PyMOL's extensive molecular visualization and analysis toolkit to add new features as deemed fit.

3.6. ProMS to Analyze Globular Proteins

Figure 6 is a demonstration of ProMS' flexibility and analytical capabilities in generating molecular surface representations. Lysozyme (PDB ID: 1LYZ) was chosen as a globular protein to illustrate how varying the different computational and visualization parameters alter the surface topology and property mapping.

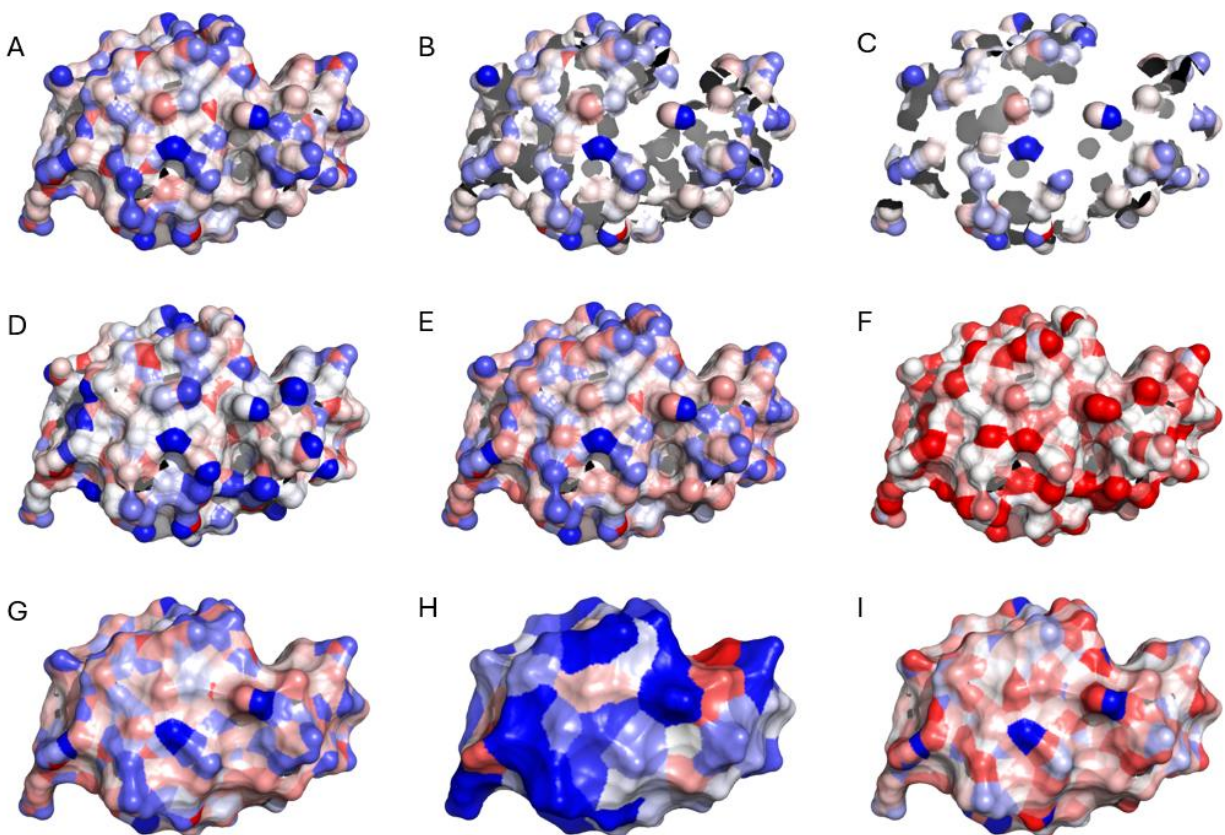


Figure 6 – Globular protein (lysozyme) visualized in ProMS using various input parameters. (A-C) Varying probe radii (1.4, 5.0, 10.0 Å, from left to right), atomic-level resolution using GAFF-derived hydrophobicities from the DGOCT scale. (D-F) Varying force fields (BIO++, GAFF, Universal, from left to right), atomic-level resolution at 1.4 Å. (G-I) Varying visualization type (atomic resolution, amino acid resolution, charge, from left to right) with a surface resolution of 5.0 Å.

In panels A-C, the increase in probe radius size results in a gradual decrease of surface detail. This is the case because smaller cavities and grooves become inaccessible as the probe size grows larger. This feature is relevant when analyzing surface interactions with molecules or solvents of different sizes. Panels D-F highlight the changes in surface characterization when using different force fields (BIO++, GAFF, and Universal, from left to right). The ability to choose different force fields (and add custom ones) enables highly customizable workflows, illustrating ProMS’ jackknife-like capabilities. Lastly, panels G-I demonstrate the different visualization modes available in ProMS (atomic resolution, amino acid resolution, charge, from left to right). As expected, the amino acid resolution shows larger amphiphilicity “patches”. This demonstrates the benefits of atomic level-

based hydrophobicity, as it allows for a more detailed surface representation. As demonstrated by **Figure 6**, ProMS' customizability is essentially limitless.

3.7. ProMS to Analyze Ion Channels

Ions, molecules with a permanent electrical charge, can enter cells only through ion channels, which selectively regulate their passage [60]. These channels typically accommodate ions with radii ranging from approximately 0.3 to 2 Å [61]. While ion conduction is often modeled as a diffusion process, understanding the selective and dynamic nature of transport requires analysis at atomic resolution.

Traditional representations, such as coloring protein surfaces by amino acid hydrophobicity, have provided broad insight into surface properties [62]. However, they lack the spatial precision needed to resolve the localized physicochemical environment experienced by permeating ions. In contrast, atomic-level hydrophobicity mapping enables detailed characterization of the channel interior and can uncover subtle features influencing selectivity and conductance.

Pore-lining atoms were identified through a five-step computational workflow. First, HOLE was used to calculate the channel's pore radius profile [63]. Atoms within a specified radial threshold (≤ 4 Å) from the pore axis were selected as candidates. Separately, msRoll computed solvent-accessible surface atoms [42]. The intersection of these two sets, interior-proximal and surface-accessible atoms, defined the pore-lining region. This subset was analyzed using ProMS to quantify atomic-level hydrophobicity and visualize the interior pore lining surface.

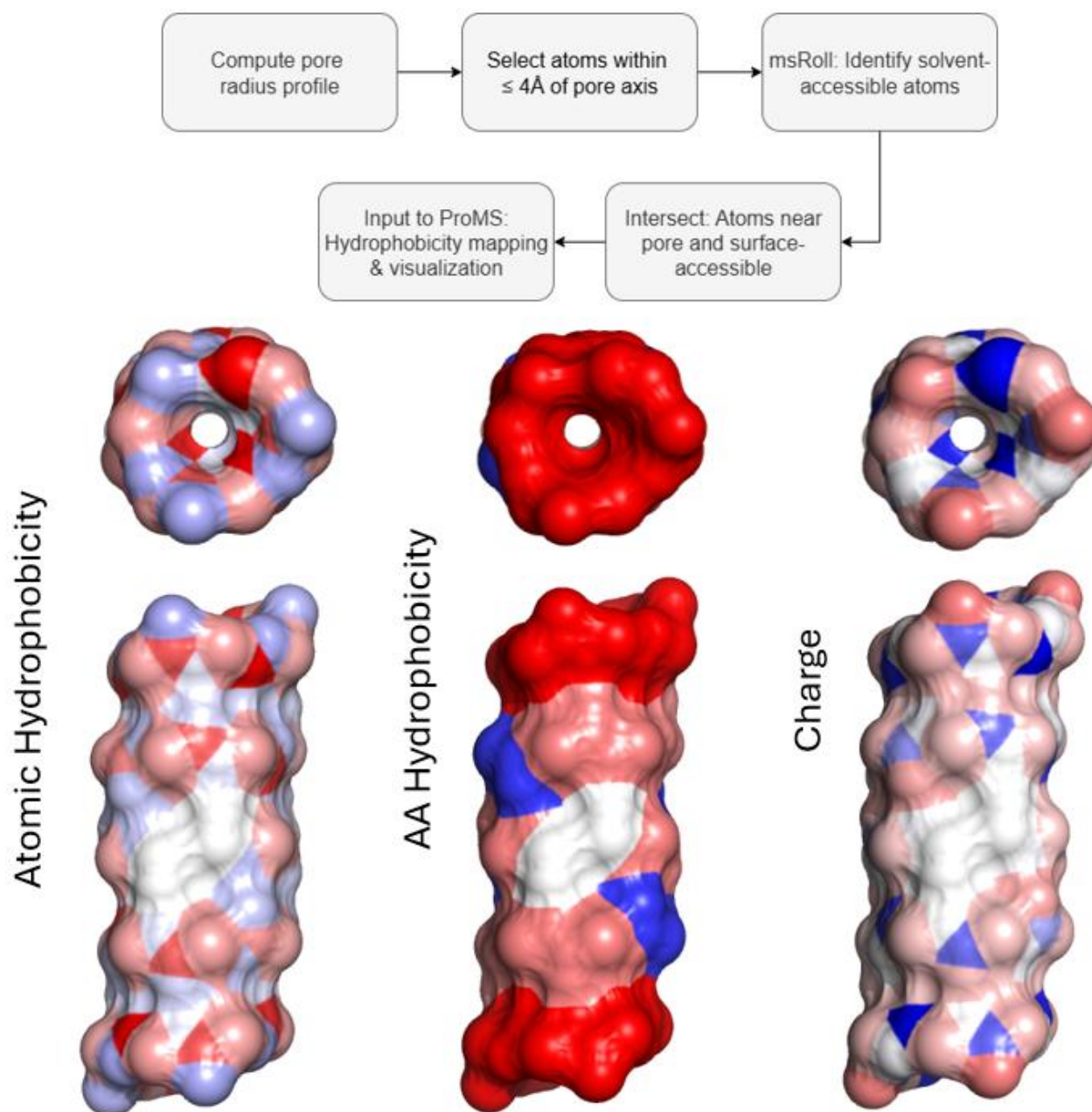


Figure 7 – Novel methodology for analyzing and visualizing ion channels. Interior of a transmembrane ion channel visualized with ProMS. From left to right, atomic resolution, amino acid resolution, and charge visualization profiles. Probe and surface resolution radius of 1.4 Å, GAFF force field derived from DGOCT hydrophobicity scale.

In **Figure 7**, the pore-lining profile of the bacterial potassium channel KcsA (PDB ID: 1GRM) is shown. The interior surface exhibits an alternating pattern of hydrophobic and hydrophilic residues

along the conduction pathway, revealing a more complex environment than is apparent with residue-level coloring alone.

3.8. ProMS to Analyze Steroids

To evaluate the utility of atomic hydrophobicity representations in small, non-peptide molecules, we selected a set of steroids: budesonide, ciclesonide, and fluticasone. Steroids represent a distinct structural class characterized by a rigid fused-ring core and a range of functional groups that affect their polarities and hydrophobicities. In particular, these three steroids, which are normally used for their anti-inflammatory properties when dealing with asthma, have shown variable responses in their potential impact on COVID-19 outcomes [64].

The structure and PDB files of the three molecules were created using Avogadro, a free cross-platform molecule editor and visualizer [65]. These files were then cleaned to fit the inputs ProMS requires and processed using the same surface generation and hydrophobicity visualization pipeline as previously discussed for proteins. The result is shown in **Figure 8**. Atom types were assigned using the universal hydrophobicity scale.

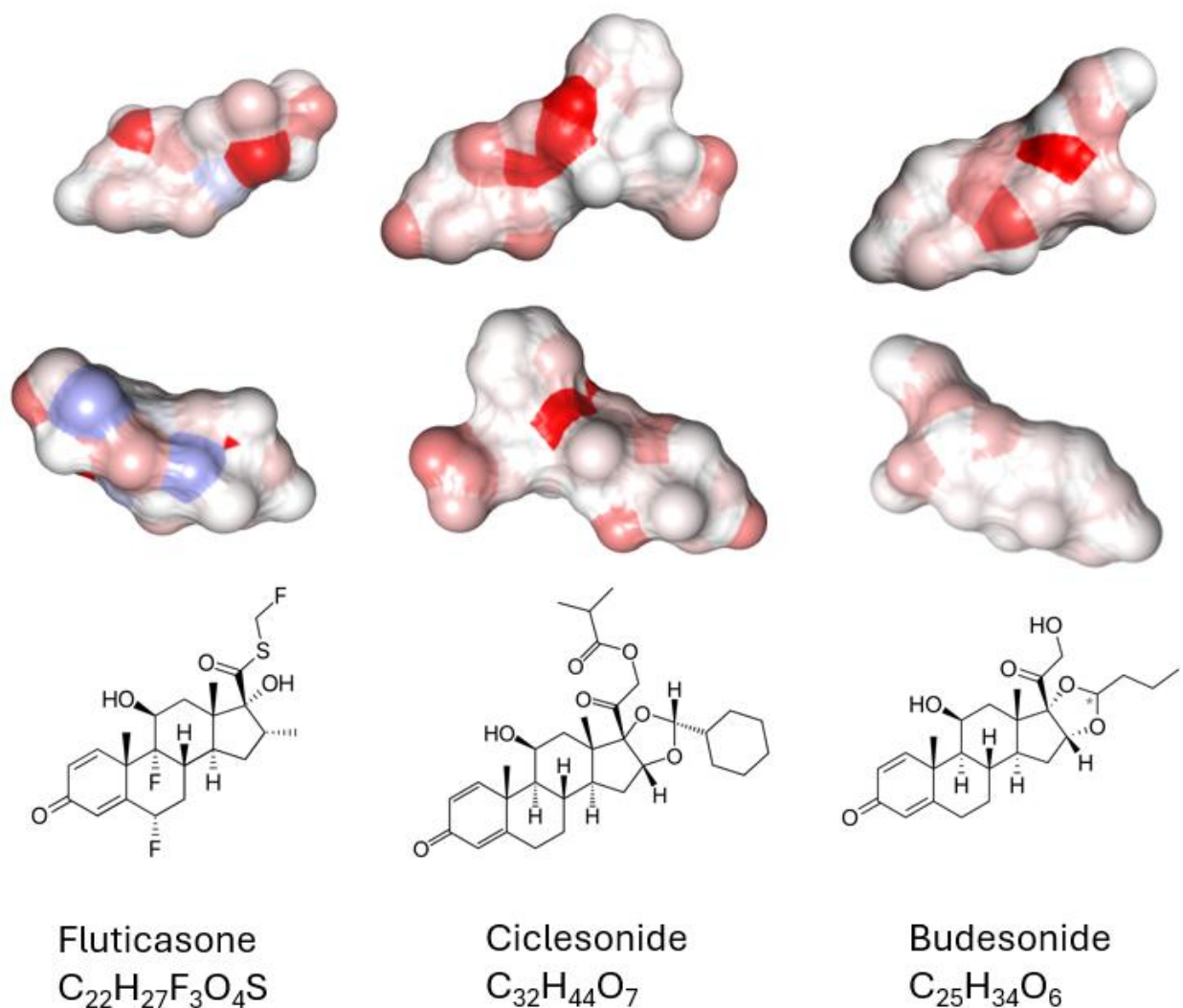


Figure 8 – Visualization of three steroids using ProMS. Steroids were chemically aligned in PyMOL and visualized with the Universal (logP) hydrophobicity schematic. Probe and surface resolution of 1.4 Å.

The primary goals for this case study were to visualize hydrophobic patches on steroid surfaces at an atomic level to potentially uncover minute differences between the three molecules and how they could explain differences in COVID-19 infection. This case study also allowed us to determine the effectiveness and generalizability of the universal atomic hydrophobicity scale and provided preliminary insights into their potential use in rational drug design or molecular interaction modeling. The molecular surfaces reveal distinct distributions of hydrophobic and hydrophilic regions corresponding to their functional group composition. In all three steroids,

hydrophobic patches (red) dominate the core scaffold, reflecting the nonpolar four-ring carbon-based backbone characteristic of steroid hormones. Meanwhile, localized hydrophilic regions (blue to white) are observed near hydroxyl, carbonyl, and fluorinated substituents, indicating potential solvent-accessible or receptor-interacting sites. The visualization highlights how subtle chemical modifications alter surface polarity and may influence bioavailability and receptor affinity.

3.9. ProMS to Analyze an Aptamer Complex

Aptamers are short oligonucleotide sequences (typically 20–80 nucleotides) that fold into unique three-dimensional structures in the presence of a target molecule, enabling high-affinity and high-specificity binding [66, 67]. The binding interface can involve polar, nonpolar, or mixed interaction modes, depending on the aptamer's sequence folding and the chemical properties of the target [68].

Hydrophobic modifications to aptamers frequently enhance their binding affinities relative to unmodified counterparts. For instance, a boronic acid–modified thymidine aptamer targeting fibrinogen exhibited a dramatic improvement in dissociation constant (k_d) from 450 nM to 3–30 nM. Similarly, base-modified pools (Uad, Ugu, Aad) selected against human β -defensin-2 demonstrated the highest binding affinity ($k_d = 6.8$ nM) when hydrophobic adenine analogs were incorporated [69].

A representative example is the SOMAmer platform, which produces aptamers with hydrophobic binding surfaces that mimic the native interface between platelet-derived growth factor-BB and its receptor. This behavior contrasts sharply with the predominantly polar interactions typical of conventional protein-binding aptamers [70].

The human immunoglobulin G (IgG) aptamer (PDB ID: 3AGV) is particularly notable for its interaction profile with the Fc region of IgG [66]. Unlike many nucleic acid–protein complexes dominated by electrostatic interactions, this aptamer engages primarily through van der Waals contacts and hydrogen bonding, making it an ideal model for visualizing subtle hydrophobic interaction surfaces in biomolecular complexes. The visualization of this protein-aptamer complex is shown in **Figure 9**.

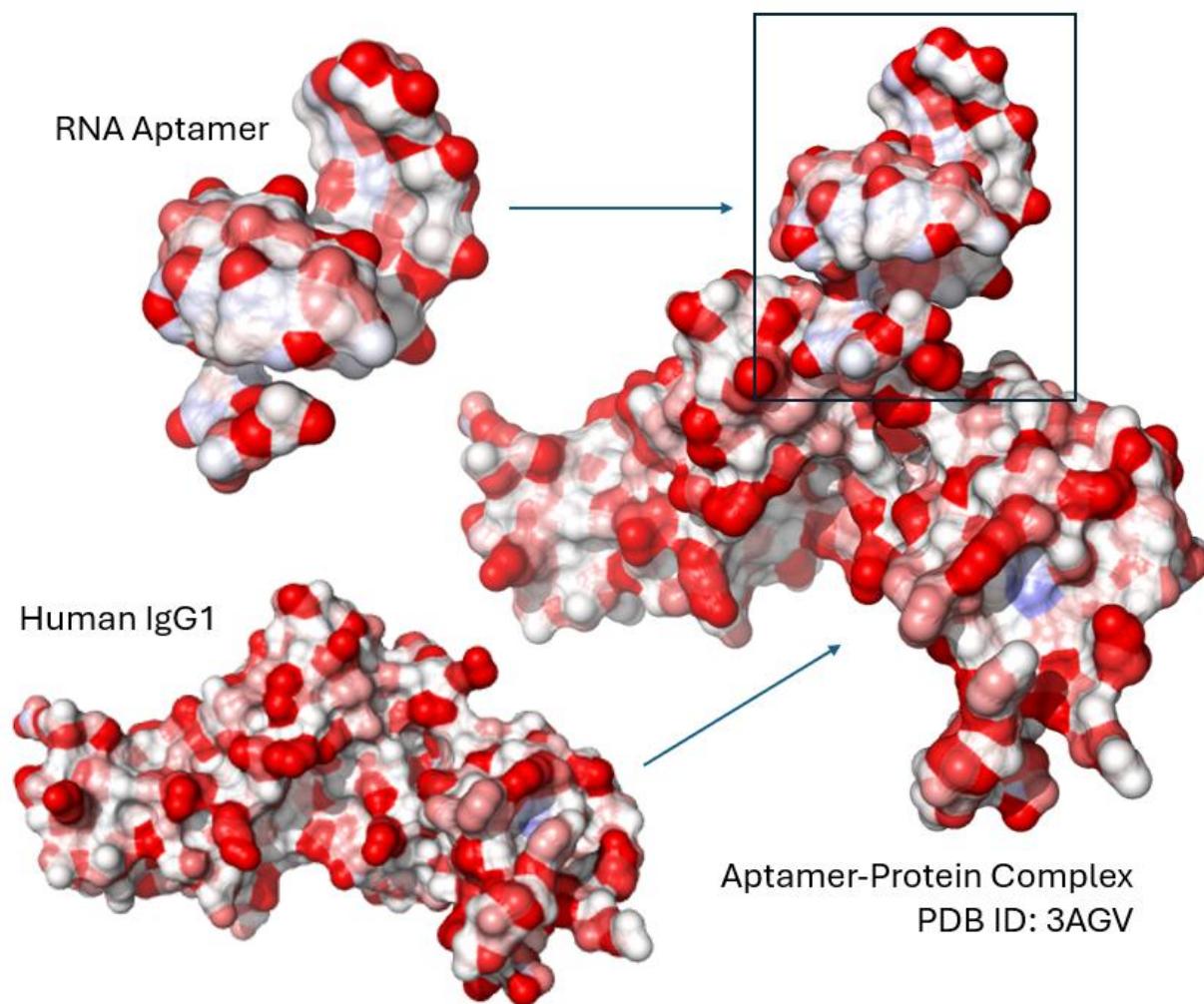


Figure 9 – Aptamer-protein complex visualization in ProMS. ProMS has been upgraded to natively support the analysis of more biomolecules, including nucleic acids. Visualized at 1.4 Å using the Universal hydrophobicity scale.

Figure 9 demonstrates the capability of ProMS to visualize complex biomolecular assemblies using the generalized Universal (logP) hydrophobicity scale. The aptamer–protein complex (PDB ID: 3AGV) is shown decomposed into its RNA aptamer and protein (human IgG) components to highlight molecular interfaces and surface polarity. Hydrophobic and hydrophilic regions are colored red and blue, respectively, based on the values derived from the universal scale. This visualization reveals complementary hydrophobic patches at the binding interface, consistent with stabilizing nonpolar interactions that contribute to aptamer–protein recognition.

3.10. BAD Protein Analysis

The proteins for which adsorption data are reported in BAD 2.0 display a remarkably wide range of structural and physicochemical properties. This diversity is immediately apparent from the visualization of their molecular surfaces (**Figure 10**), generated using a probe radius of 20 Å. Molecular surface visualization and the quantification of geometric and physicochemical descriptors were performed using ProMS. While interfacing a protein with an ideally flat surface is mathematically equivalent to probing the molecular surface with a probe of infinite radius, real experimental surfaces exhibit nanoscale rugosities, corresponding to significantly smaller probe sizes. The upper threshold of 20 Å was selected for two reasons. First, even nominally flat surfaces retain measurable roughness; for example, glass surfaces polished to reduce roughness still exhibit features on the order of 2 nm, and freshly cleaved mica has characteristic features around 25 Å [71, 72]. Second, prior work has shown that physicochemical properties quantified on protein molecular surfaces become relatively insensitive to increases in probe radius beyond 20 Å, indicating that this value captures the relevant surface features while maintaining computational efficiency [22].

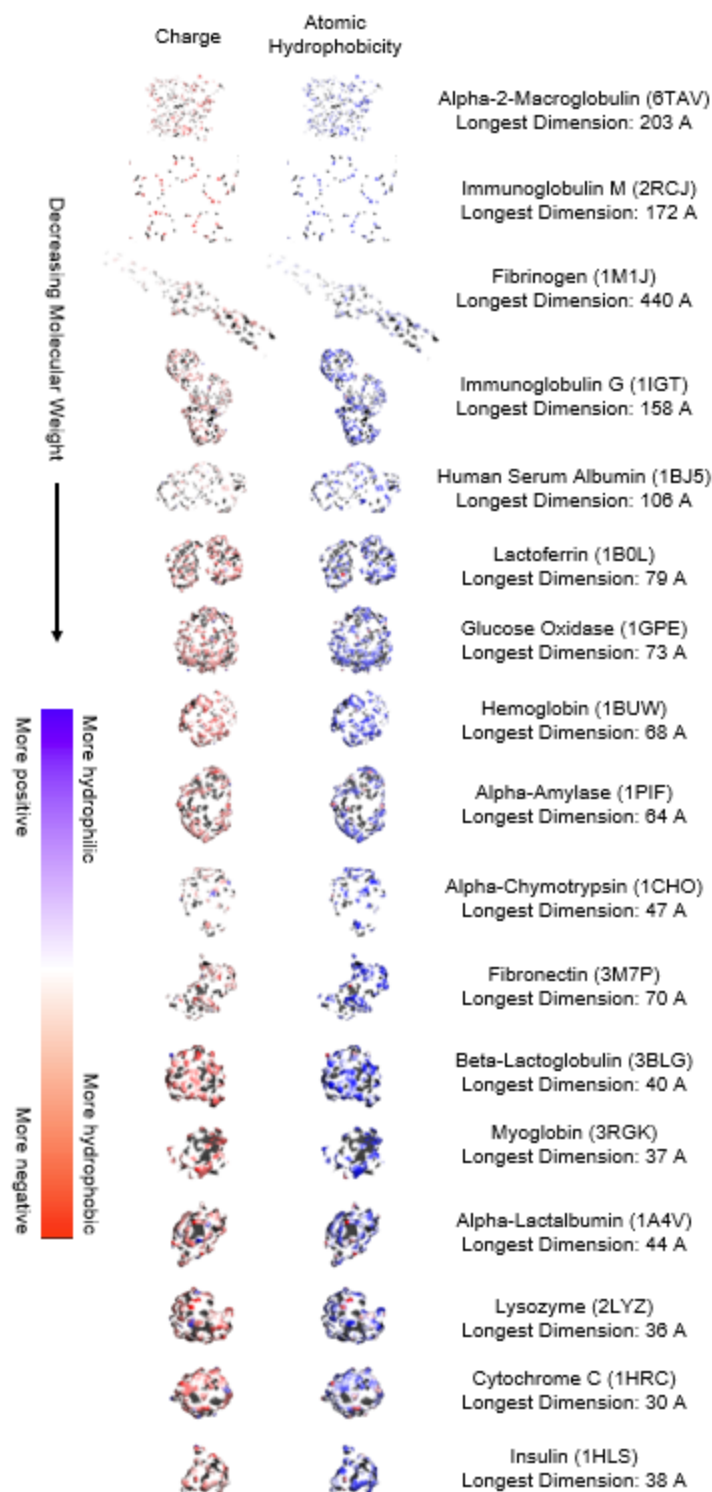


Figure 10 – Molecular surfaces of the proteins present in BAD2.0. Probed with a 20 Å radius and visualized with ProMS.

Figure 11 shows the distribution of some of the key descriptors for the 865 BAD entries. While some of the histograms simply report values obtained from literature (contact angle, surface protein concentration, etc.), further insight into protein adsorption behavior was obtained by examining secondary and tertiary structural descriptors. For example, the ratio of residues in α -helices, β -sheets, or connected by disulfide bonds to the total number of residues provides an estimate of protein rigidity. As shown in **Figure 11-j**, the distribution of this ratio indicates that approximately 54% of proteins in BAD 2.0 are relatively rigid, whereas a distinct cluster (42%) corresponds to more flexible proteins.

Additional information was gleaned from the ratios of molecular surfaces probed at different radii. The ratio of the surface area probed with a large radius (e.g., 20 Å) to that probed with a small radius (e.g., 1.4 Å, approximating a water molecule; molecular surfaces shown in **Figure 12**) provides a measure of protein compactness. The distribution of these ratios (**Figure 11- k**) suggests that 57% of proteins display complex geometries with substantial concave internal volumes—regions unlikely to contact adsorbing surfaces—while a separate cluster exhibits more convex or smoothly contoured geometries (e.g., spherical or ellipsoidal proteins).

Finally, the extent to which the hydrophobic core becomes exposed when probed with large radii offers a proxy for the propensity of a protein to adsorb onto hydrophobic surfaces. The distribution of hydrophobic-core exposure ratios (**Figure 11- l**) indicates that 56% of proteins expose less than 10% of their hydrophobic core to small probes (e.g., water) compared to flat adsorbing surfaces. Smaller clusters show higher levels of hydrophobic-core exposure, suggesting greater variability in adsorption behavior across the dataset.

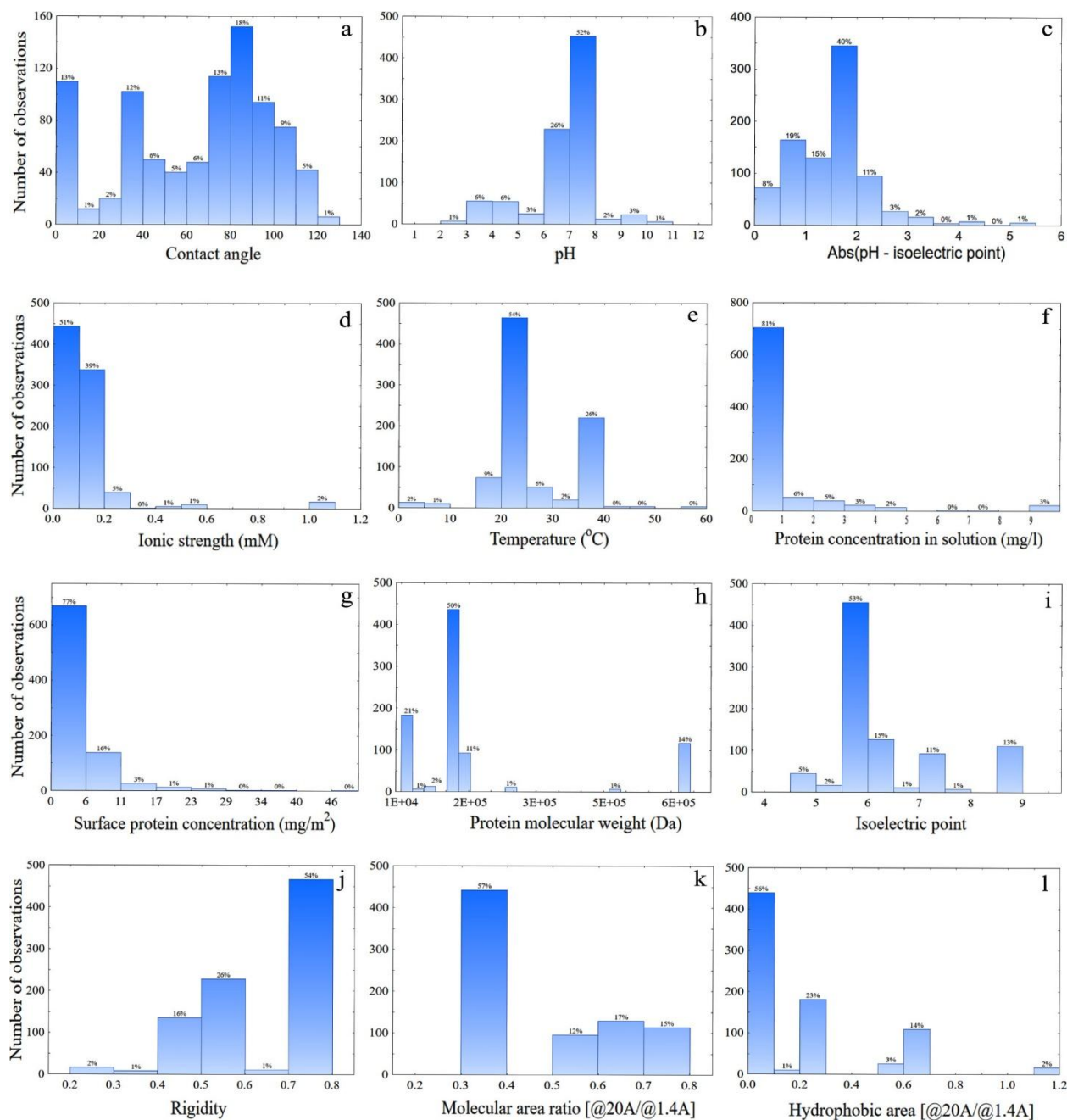


Figure 11 - Distribution of the protein adsorption descriptors in BAD 2.0. (a) adsorbing surface hydrophobicity, (b) pH of the carrier fluid, (c) difference between the pH and the protein isoelectric point (absolute values), (d) ionic strength of the carrier fluid, (e) temperature of the fluid, (f) concentration in solution, (g) concentration on surface, (h) protein molecular weight, (i) isoelectric point of the protein, (j) protein rigidity (ratio of residues in α helices, β sheets, and S-S bonds and the total residues), (k) ratio of molecular areas quantified using probes with radii of 1.4 and 20 Å, respectively, and (l) ratio hydrophobic areas using 1.4 and 20 Å probes.

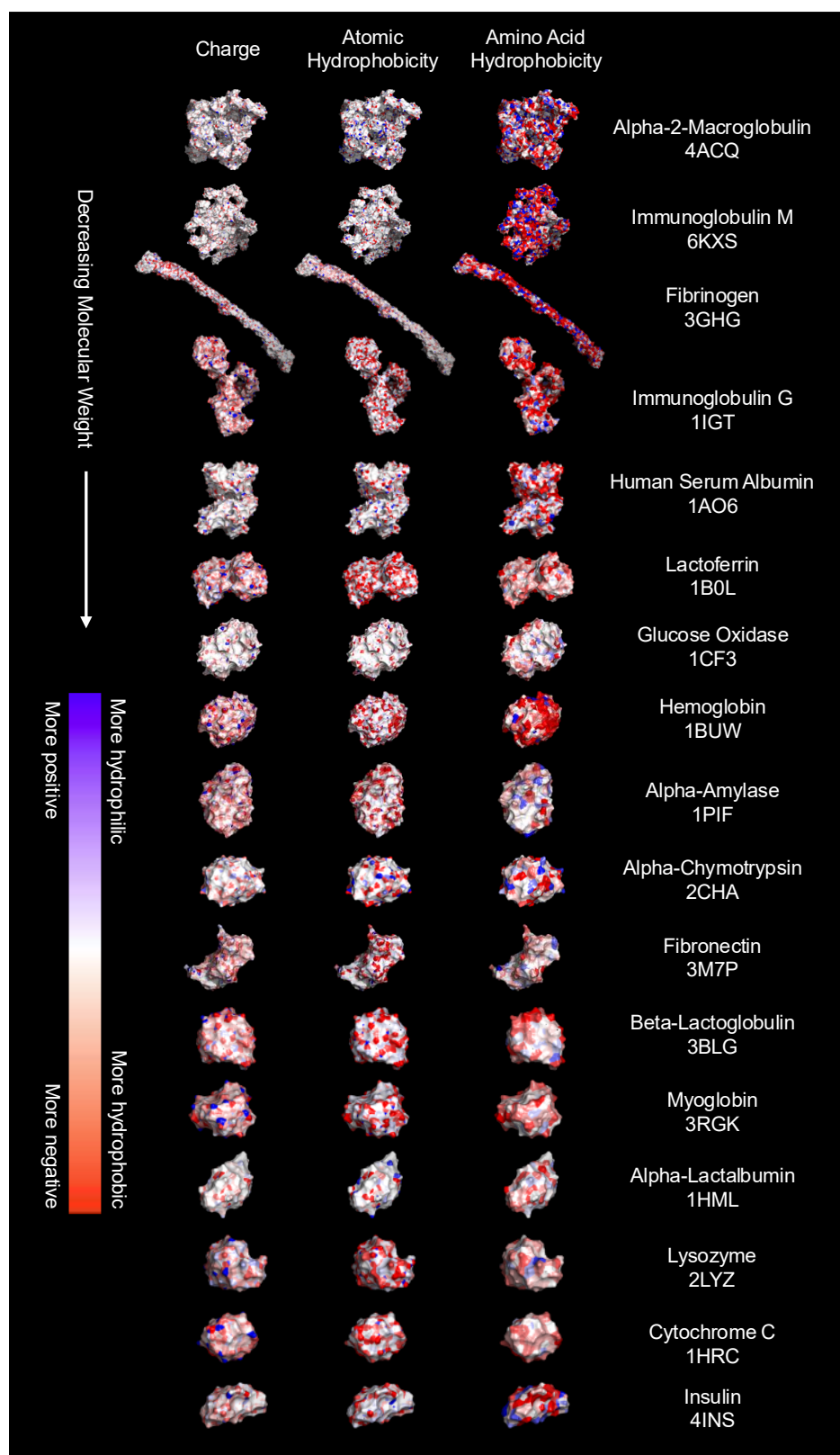


Figure 12 – Molecular surfaces of the BAD2.0 proteins. Probed with a 1.4 Å radius and visualized with ProMS.

3.11. BAD Machine Learning Analysis

Inspection of BAD 2.0 reveals substantial variability and nonuniform distributions among the many parameters that modulate protein adsorption on solid surfaces. Given this complexity, the rapid development of artificial intelligence approaches suggests that such variability may be addressed through fully empirical, data-driven models capable of capturing nonlinear relationships without prior mechanistic assumptions. In this case, machine learning algorithms were applied to BAD 2.0 to determine the relative importance of (i) operational variables and (ii) protein-derived surface descriptors on the adsorbed protein mass (**Figure 13-a** and **13-b**, respectively).

The operational variables shown in **Figure 13-a** include (i) the protein concentration in solution (represented as the inverse value of monolayers), (ii) the contact angle of the adsorbing surface, (iii) ionic strength, (iv) temperature, (v) pH of the carrier fluid, and (vi) the difference between the solution pH and the isoelectric point (IP) of the protein. Notably, pH appears twice in this list because it influences adsorption through distinct physical mechanisms, which justify its representation as two separate variables. First, although the zeta potential of adsorbing surfaces can significantly influence protein adsorption, especially on hydrophilic substrates, this parameter is rarely reported in literature. Nevertheless, zeta potential varies almost linearly with solution pH, making pH a reasonable surrogate [73]. Second, the pH–IP difference reports the net charge state of the protein: when $\text{pH} = \text{IP}$, positive and negative charges balance, and the protein is electrostatically neutral.

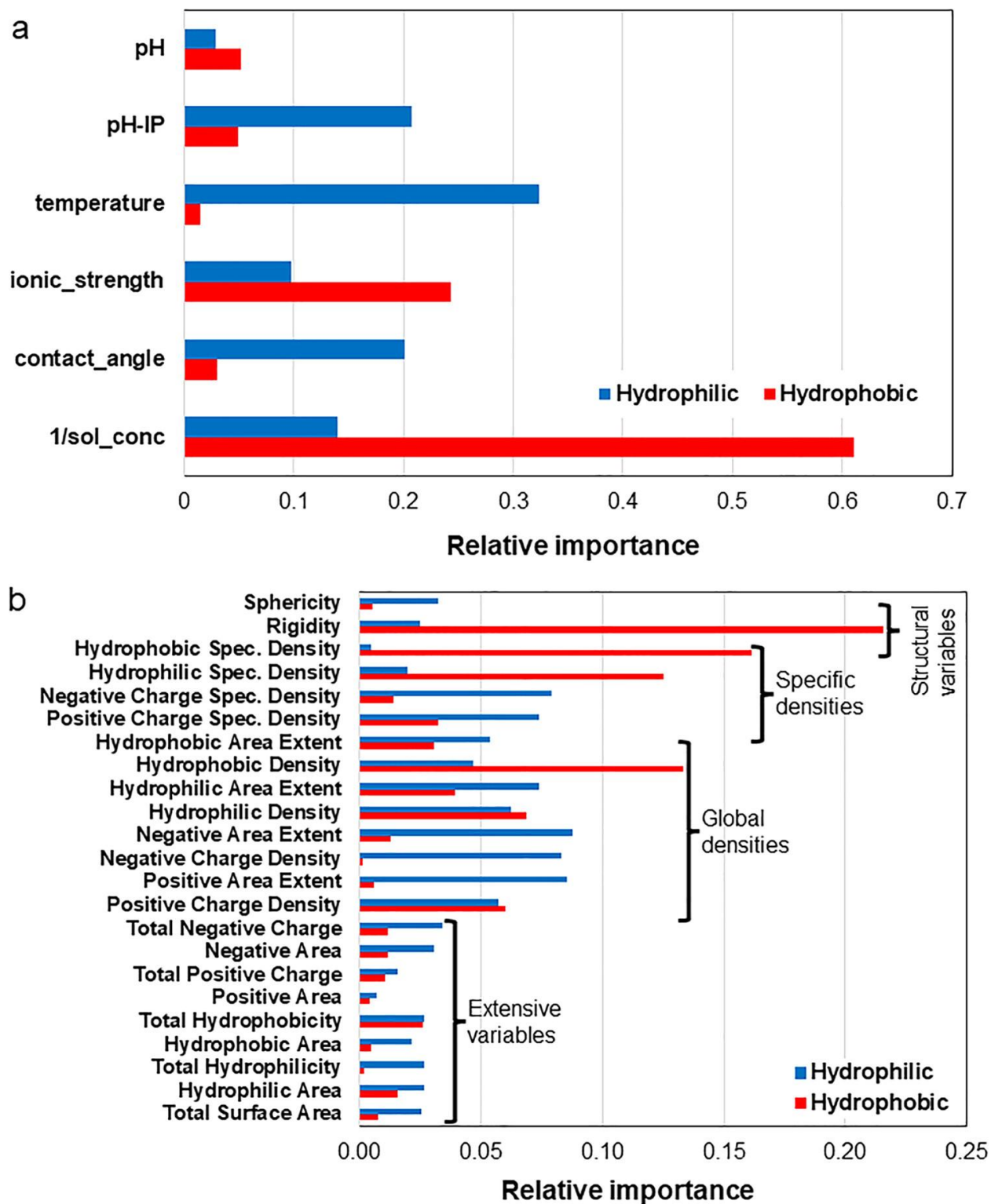


Figure 13 – Relative importance of protein adsorption predictors, given by ridge regression algorithm. A) Operational parameters. B) Physicochemical properties of the proteins quantified on their molecular surface.

The inspection of the importance of variables showed that their impact is considerably different for the adsorption on hydrophobic surfaces than that on hydrophilic ones. Indeed, for protein adsorption on hydrophobic surfaces, the importance of the contact angle (rather unexpectedly) appeared to be negligible. In contrast, protein adsorption on hydrophilic surfaces appeared to be strongly modulated by pH, pH-IP, and the contact angle of the adsorbing surface. For both surfaces, the protein concentration in solution and the fluid temperature appear to be strong modulators, albeit more so for adsorption on hydrophilic surfaces, whereas ionic strength does not appear to be important.

The protein-derived predictors modulating protein adsorption comprise the total surface area, rigidity, sphericity, hydrophobic area (from which the hydrophilic area can be derived by substitution), total hydrophobicity (from which hydrophobic density and hydrophobic specific density can be calculated), negative area, and total negative charge. An inspection of the relative importance of these predictors, as shown in **Figure 13-b**, revealed that the extensive predictors, that is, variables depending on the molecular weight of the protein, presented substantially lower values than the intensive predictors, that is, variables that do not depend on the size of the protein. Focusing on the importance of intensive variables, it can be observed that protein adsorption on hydrophilic surfaces is more dependent on global densities, such as area extents, i.e., ratios of the area of positive and negative charges, respectively, and hydrophilic and hydrophobic areas, respectively, reported to the total molecular area of the protein, and property densities, i.e., total positive charges, negative charges, total hydrophilicity, and total hydrophobicity, reported to the total area of the protein. Conversely, protein adsorption on hydrophobic surfaces is modulated by specific densities, i.e., total positive charges, negative charges, total hydrophilicity, and total hydrophobicity, reported to their respective areas on the molecular surface of the protein.

The considerable difference in the relative importance of the operational and protein-derived predictors for protein adsorption on hydrophobic and hydrophilic surfaces suggests that the mechanism of protein adsorption on these two types of surfaces presents considerable differences, suggesting separate analysis. However, while machine learning analysis provided important insights into protein adsorption, it could not provide reasonably predictive correlations (i.e., low correlation coefficients). Indeed, the predictive power, and thus the benefit, of machine learning decreases rapidly with the distance away from the training data due to the lack of explicit relationships between inputs and outputs. This decrease in predicting power is amplified by the uneven distribution of the data points in the training set, as is the case for very different proteins present in BAD, as well as by the nonlinear relationships between some of its input and output parameters. Consequently, in this analysis, machine learning was solely used to infer the most important adsorption predictors rather than to make protein adsorption predictions itself.

While recent advances in machine learning have enabled highly accurate, data-driven predictions in many areas of science and engineering, such approaches often operate as black boxes, offering limited insight into the underlying physicochemical driving forces. In contrast, the objective of the present work was not to maximize predictive accuracy at all costs, but rather to use machine learning as an exploratory and interpretative tool to interrogate the relative importance of molecular, surface, and solution-phase descriptors. Despite the modest overall predictive power, the machine learning analyses performed here revealed clear, physically meaningful trends, such as the distinct roles of charge density, hydrophobicity, and surface wettability on hydrophilic versus hydrophobic substrates. Machine learning can serve as a powerful framework for hypothesis generation, variable ranking, and model refinement, guiding the development of semiempirical or mechanistic descriptions that more faithfully reflect the underlying physics. By

explicitly prioritizing understanding over blind prediction, this work contributes to a more interpretable and scientifically grounded use of machine learning in the complex world of biological sciences.

3.12. Perspectives, Limitations, and Future Work

The development of ProMS and its application across a range of biomolecular systems demonstrate the promise of atomic-level surface analysis for advancing our understanding of molecular recognition, adsorption phenomena, and structure–function relationships. The current work establishes ProMS as a flexible and powerful platform, yet opportunities for further development remain.

Despite its versatility, ProMS in its present form has its limitations. First, ProMS relies on static molecular structures and therefore does not account for conformational dynamics that may expose or occlude molecular surface features. Proteins with substantial flexibility or induced-fit behaviors may therefore be imperfectly captured. Second, ProMS uses a limited set of physicochemical descriptors, limited to hydrophobicity, electrostatics, and geometric features. While these descriptors are foundational, they cannot fully represent more complex factors such as hydrogen-bonding propensity, polarizability, solvation dynamics, or pH-dependent protonation equilibria. Third, the case studies demonstrate breadth of applicability but remain qualitative in nature; a more systematic, quantitative benchmarking across standardized datasets is needed to fully validate the tool’s predictive power.

With respect to the adsorption analysis, the limitations of BAD 2.0 also influence the analysis performed here. The uneven distribution of proteins, the sparsity of certain experimental descriptors (notably zeta potential), and inconsistencies in measurement protocols introduce noise

that limits the accuracy of machine learning approaches. Furthermore, although ML provided insight into the relative importance of variables, the predictive performance remained inaccurate and quite frankly, unreliable, due to data heterogeneity and the absence of explicit mechanistic relationships.

A major opportunity for future work lies in expanding the physicochemical descriptors available in ProMS. Coupling ProMS with molecular dynamics (MD) simulations could allow ensemble-averaged surfaces to be computed, capturing transient or cryptic binding features inaccessible to static analyses.

The case studies presented here form a foundation for more extensive applications. Future work could involve constructing curated benchmark datasets to quantitatively assess the performance of ProMS-derived surface metrics. For example, proteins with experimentally validated interaction hotspots or drug-receptor systems with known affinity profiles would be good candidates for a more thorough validation “litmus-test”. Analyses of ion channels may be extended to understand gating or permeation mechanisms, while steroid comparison studies could be expanded to larger pharmacophore libraries to examine how subtle changes in amphiphilicity influence receptor specificity. The aptamer–protein case study suggests additional applications in nucleic-acid–based therapeutics, particularly in binding prediction and interface characterization.

The integration of ProMS with BAD 2.0 also opens new opportunities in semiempirical modeling and data-driven surface analysis. Future versions of ProMS could automate the extraction of advanced descriptors for thousands of proteins, enabling large-scale screening of adsorption tendencies on engineered surface chemistries. Indeed, ProMS could potentially have a promising future in drug discovery and biomolecular engineering applications, especially as it continues to evolve as a software package.

From a computational standpoint, ProMS stands to benefit from expanded interoperability and performance enhancements. Future upgrades may include GPU-accelerated routines, improved batch processing of large data sets, and dedicated web servers for additional computing power. Finally, long-term impact will depend on the establishment of reproducible workflows and standardized validation datasets. Continued community contributions to BAD 2.0, combined with transparent benchmarking and open-source releases of ProMS, will support its adoption as a central platform for studying and engineering biomolecular surfaces.

4. Conclusion

The overarching aim of this thesis was to advance the quantitative and visual analysis of biomolecular surfaces and to apply these tools toward a deeper understanding of protein adsorption. To achieve this, the research addressed several interconnected objectives: the development of ProMS, a modular Python-based platform for atomic-level molecular surface analysis; the application of ProMS across representative molecular systems to demonstrate its capabilities; the expansion and refinement of the Biomolecular Adsorption Database (BAD 2.0); and the use of machine learning methods to identify the operational and molecular-surface variables most strongly associated with protein adsorption. Collectively, these objectives were successfully met, and together they contribute to a more integrated understanding of biomolecular behavior.

First, the development of ProMS represents a significant methodological contribution. ProMS unifies Connolly analytical surface calculations with force-field-specific and universal logP-based atomic hydrophobicity scales, enabling surface amphiphilicity, charge, and geometry to be analyzed in a cohesive, customizable framework. By leveraging PyMOL's API and modular Python libraries, ProMS provides an accessible and extensible system for surface computation at both atomic and residue resolutions. Through case studies involving globular proteins, ion channels, steroids, and aptamer–protein complexes, the tool demonstrated its versatility and ability to illuminate surface features relevant to molecular recognition, ligand specificity, and topographical complementarity. These results illustrate how ProMS can bridge structural biology and molecular design by revealing descriptors that are invisible in at lower-resolution residue-based representations.

Second, the upgrade of BAD 2.0 established an expanded and rigorously curated dataset for studying protein adsorption. By incorporating 15 years of newly reported data, adding consistent metadata fields, and linking protein entries to structural identifiers in the Protein Data Bank, BAD 2.0 provides a significantly improved foundation for exploring the complex interplay of molecular properties, surface chemistry, and solution conditions. Integration with ProMS enabled the addition of analytically derived surface descriptors, such as hydration properties, charge distributions, compactness metrics, hydrophobic-core exposure, that had not been previously quantified at scale. This systematic surface characterization represents a meaningful step toward unifying experimental and computational perspectives on adsorption.

Third, the application of machine learning to BAD 2.0 revealed the relative importance of operational parameters and protein-derived descriptors across hydrophobic and hydrophilic surfaces. Although predictive performance was limited by data heterogeneity and uneven sampling, the analysis provided insight into the distinct mechanisms governing adsorption in different interfacial environments. For hydrophilic surfaces, adsorption was dominated by global charge densities, pH-dependent effects, and wettability. For hydrophobic surfaces, more localized, domain-specific descriptors, as in specific charge densities and hydrophobic patches, were the strongest determinants.

Taken together, this thesis highlights the central role of molecular surfaces in biomolecular recognition and adsorption and provides both tools and datasets for advancing future research. ProMS establishes a computational framework capable of supporting mechanistic studies, large-scale screening, or rational design workflows. BAD 2.0 provides an expanded empirical foundation for studying adsorption trends across diverse proteins. The machine learning analysis,

while not yielding universally predictive models, clarifies which physicochemical variables warrant emphasis in future modeling efforts.

The implications of this work extend across structural biology, biotechnology, and materials science. Atomic-level surface descriptors can inform the design of protein therapeutics, biosensor interfaces, surface-immobilized enzymes, and anti-fouling materials. The integration of experimental and computational hydrophobicity scales offers new opportunities for understanding solvation and interface behavior. Most importantly, the combined contributions of this thesis establish a pathway toward surface-centered modeling of biomolecular interactions, which is an approach that more accurately reflects the physics of how proteins encounter, recognize, and bind partners in complex environments.

In closing, this thesis provides a foundation for future advances in molecular surface analysis, adsorption modeling, and interface engineering. The tools, databases, and insights developed herein form a cohesive platform for continuing work in this area, and open the door to more predictive, mechanistic, and application-driven research on biomolecular surfaces.

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Appendix A: Protein Adsorption on Solid Surfaces: Data Mining, Database, Molecular Surface-Derived Properties, and Semiempirical Relationships